

AI & Full Stack Interview Questions with Answers

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1. Tell me about yourself.

Answer:

I am an AI Developer and Full Stack Engineer with experience in building AI automation systems using Python, Flask, OpenAI API, Pinecone, and workflow automation tools like Zapier and Make.com. I have worked on AI-powered Slackbots, automated recruitment systems, and responsive web applications. I also have knowledge of cybersecurity, REST APIs, and cloud deployment platforms like Render and PythonAnywhere.

2. What is Flask and why did you use it?

Answer:

Flask is a lightweight Python web framework used for building web applications and APIs.

Why used:

- Easy to develop REST APIs
- Lightweight and flexible
- Good for AI and automation projects
- Easy integration with OpenAI APIs

Example:

I used Flask in the AI Resume-Matching Slackbot project to handle API requests and chatbot communication.

3. What is Pinecone?

Answer:

Pinecone is a vector database used for storing embeddings and performing similarity searches.

Use Case:

- Resumes were converted into embeddings
- Stored in Pinecone

- Compared with job descriptions
- Best candidates were matched automatically

4. Explain how AI Resume Matching works.

Answer:

1. Upload resumes
2. Convert text into embeddings using OpenAI
3. Store embeddings in Pinecone
4. Compare embeddings with job requirements
5. Return candidates with highest similarity scores

Benefits:

- Faster recruitment
- Better candidate filtering
- Reduces manual work

5. What is REST API?

Answer:

REST API is a communication method between applications using HTTP methods like:

- GET
- POST
- PUT
- DELETE

Example:

Frontend sends request → Flask API processes data → Returns response in JSON format.

6. Difference between Zapier and Make.com?

Zapier:

- Easy for beginners
- Simple workflows
- Less visual
- Faster setup

Make.com:

- More advanced automation
- Complex workflows
- Highly visual interface
- Better customization

7. What is OAuth 2.0?

Answer:

OAuth 2.0 is an authentication protocol used for secure login and API authorization.

Example:

“Login with Google” uses OAuth 2.0.

Benefits:

- Secure authentication
- No password sharing
- Token-based access

8. What is Ngrok and why is it used?

Answer:

Ngrok creates a secure public URL for local servers.

Use:

During development, local Flask applications can be connected with external services like Slack or webhooks.

9. Explain your Slackbot project.

Answer:

I developed an AI-powered Slackbot using:

- Python
- Flask
- OpenAI API
- Pinecone

Features:

- Resume analysis
- Candidate matching
- AI responses
- Recruitment automation

Result:

Improved hiring efficiency and reduced manual screening.

10. What are Webhooks?

Answer:

Webhooks are automated messages sent from one application to another when an event occurs.

Example:

When a form is submitted:

- Webhook sends data automatically
- Another application receives and processes it

11. What is OpenAI API?

Answer:

OpenAI API allows developers to integrate AI models like GPT into applications.

Uses:

- Chatbots
- Text generation
- Resume analysis
- AI assistants

12. What is Vector Similarity Search?

Answer:

It is a technique used to find similar data based on embeddings.

Example:

If two resumes have similar skills, vector similarity search identifies them as related.

13. What security practices did you use in your projects?

Answer:

- OAuth 2.0 authentication
- SSL encryption
- Input sanitization
- Rate limiting
- Secure REST APIs
- OWASP security principles

14. What is Docker?

Answer:

Docker is a platform used to package applications and dependencies into containers.

Benefits:

- Easy deployment

- Same environment everywhere
- Better scalability

15. What are embeddings in AI?

Answer:

Embeddings are numerical vector representations of text generated by AI models.

Use:

They help AI systems understand semantic meaning and similarity between texts.

16. What challenges did you face in your AI project?

Answer:

- API integration issues
- Handling large resume data
- Improving matching accuracy
- Managing webhook connections

Solution:

I optimized prompts, improved vector search, and implemented better API handling.

17. Why should we hire you?

Answer:

You should hire me because I have practical experience in:

- AI automation
- Full stack development
- API integration
- Workflow automation
- Cybersecurity basics

I can build scalable and efficient AI-based solutions while also understanding business requirements.

18. What is Generative AI?

Answer:

Generative AI is AI that creates new content such as:

- Text
- Images
- Code
- Audio

Example:

GPT models generate human-like responses.

19. What is the difference between frontend and backend?

Frontend:

- User interface
- HTML/CSS/JS
- Visible to users

Backend:

- Server-side logic
- Python/Node.js
- Processes data

20. Explain your automation workflow project.

Answer:

I created automated workflows using Zapier and Make.com to:

- Process PDF data
- Connect SaaS applications

- Reduce manual data entry
- Improve processing accuracy

This saved time and increased efficiency.

HR Questions

Why do you want this job?

Answer:

I want this job because it aligns with my passion for AI, automation, and full stack development. I want to contribute my skills while also learning from experienced professionals.

What are your strengths?

Answer:

- Problem-solving
- Fast learning
- Communication
- Team collaboration
- Adaptability

What is your future goal?

Answer:

My goal is to become an expert AI Engineer and build scalable AI solutions that solve real-world business problems.